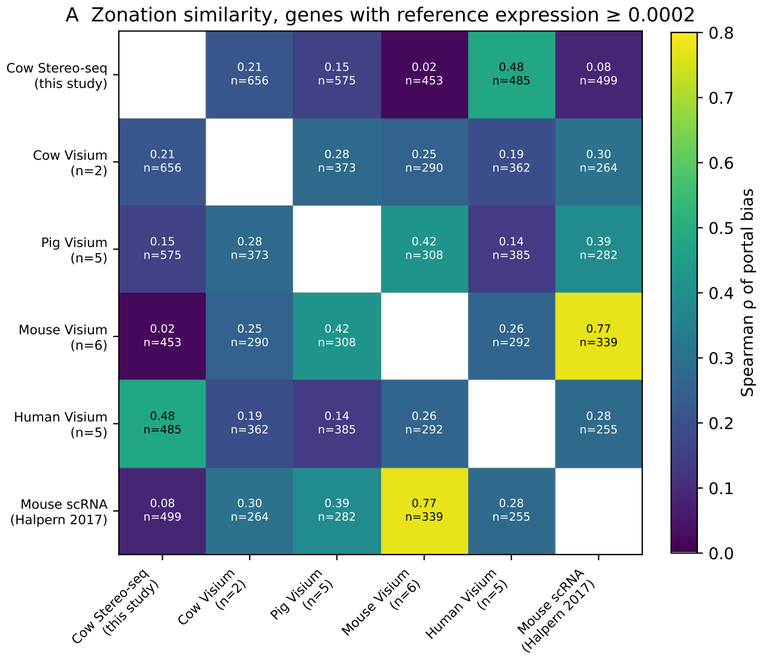
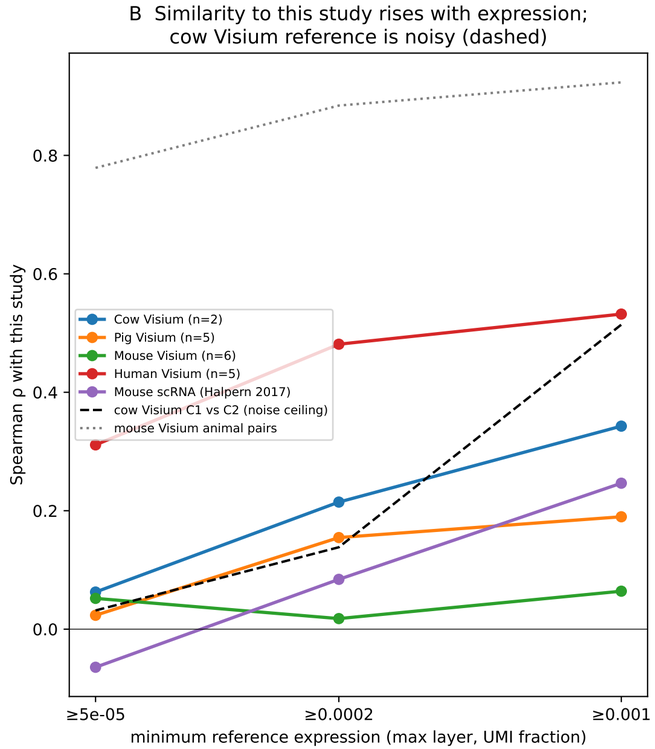
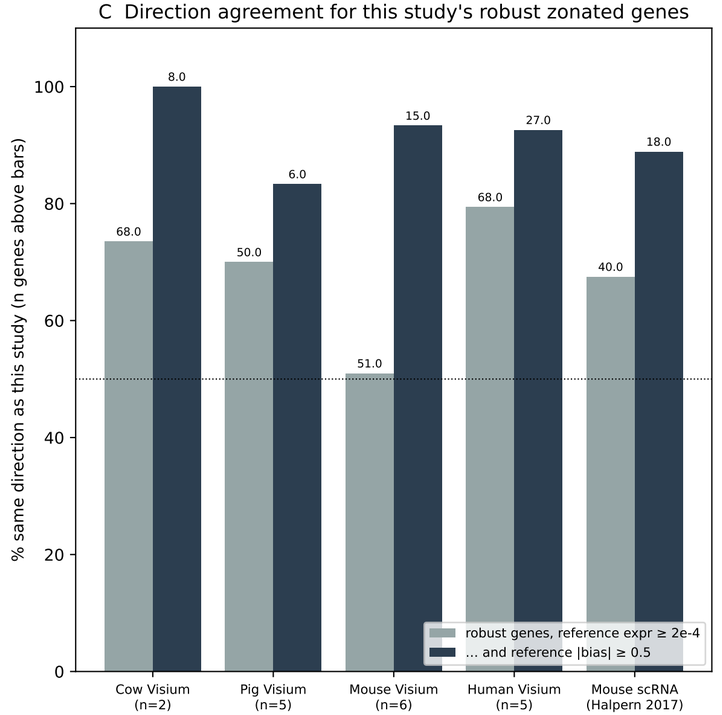
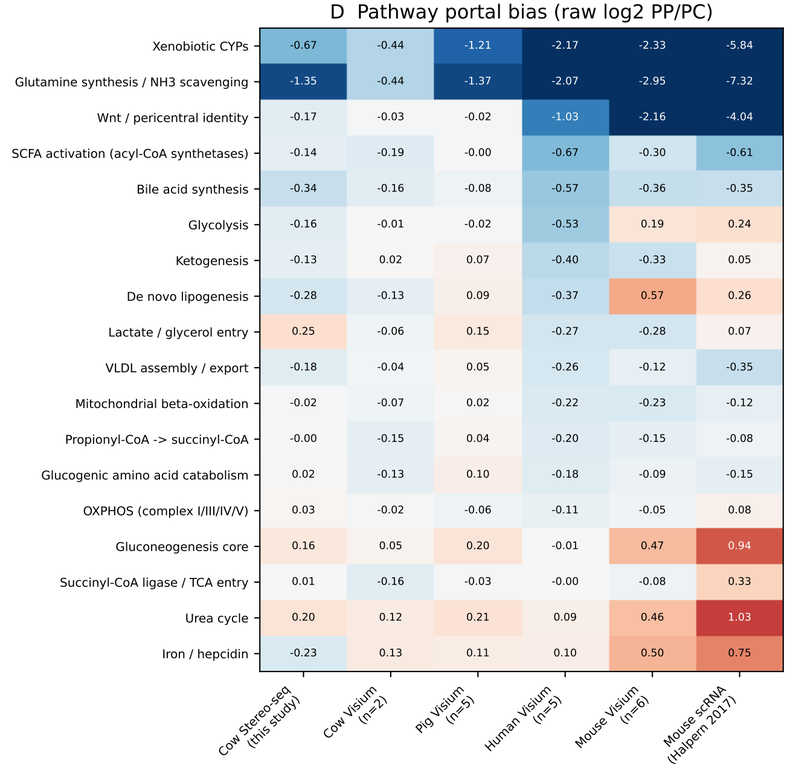
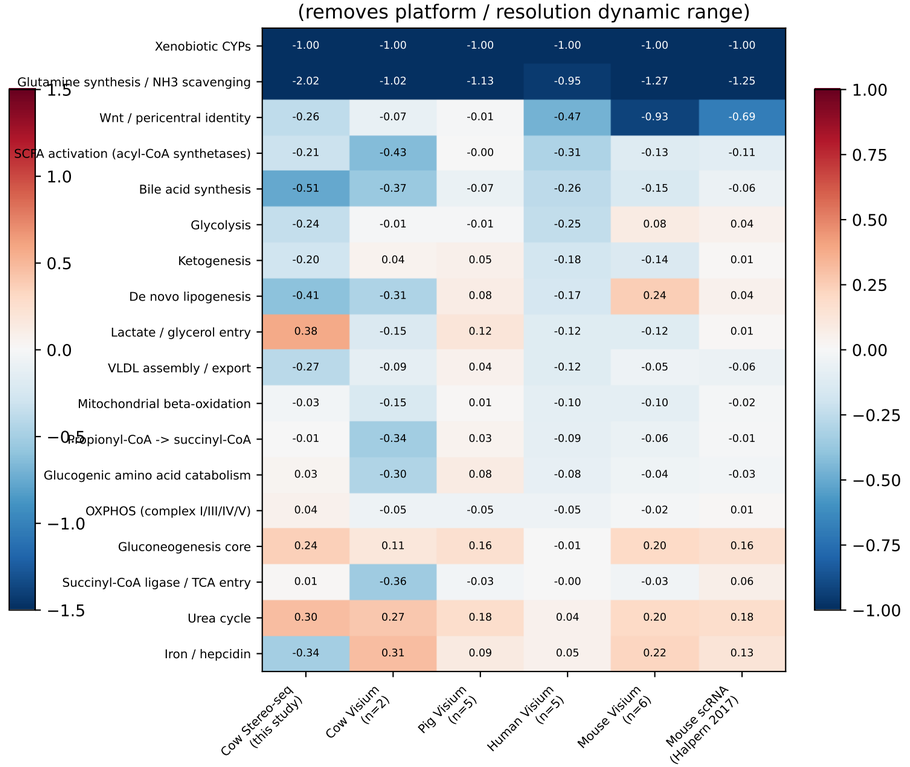
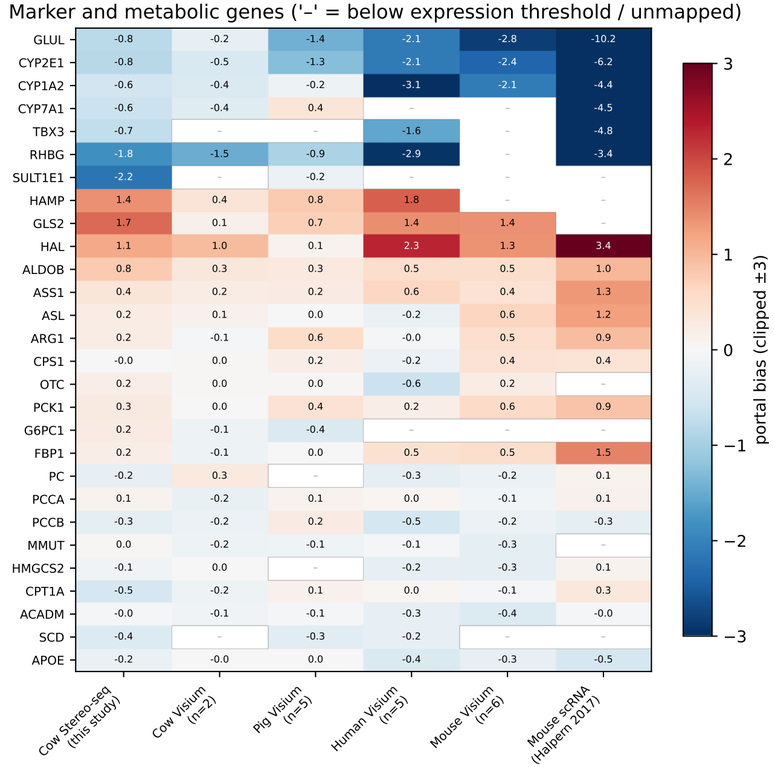

A Zonation similarity, genes with reference expression ≥ 0.0002
Cow Stereo-seq (this study)
Cow Visium (n=2)
Pig Visium (n=5)
Mouse Visium (n=6)
Human Visium (n=5)
Mouse scRNA (Halpern 2017)
0.21 n=656
0.15 n=575
0.02 n=453
0.48 n=485
0.08 n=499
0.28 n=373
0.25 n=290
0.19 n=362
0.30 n=264
0.42 n=308
0.14 n=385
0.39 n=282
0.26 n=292
0.77 n=339
0.28 n=255
Spearman ρ of portal bias
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8


B Similarity to this study rises with expression;
cow Visium reference is noisy (dashed)
Spearman ρ with this study
0.0
0.2
0.4
0.6
0.8
Cow Visium (n=2)
Pig Visium (n=5)
Mouse Visium (n=6)
Human Visium (n=5)
Mouse scRNA (Halpern 2017)
cow Visium C1 vs C2 (noise ceiling)
mouse Visium animal pairs
≥5e-05
≥0.0002
≥0.001
minimum reference expression (max layer, UMI fraction)


C Direction agreement for this study's robust zonated genes
% same direction as this study (n genes above bars)
0
20
40
60
80
100
68.0
8.0
50.0
6.0
51.0
15.0
68.0
27.0
40.0
18.0
robust genes, reference expr ≥ 2e-4
... and reference |bias| ≥ 0.5
Cow Visium (n=2)
Pig Visium (n=5)
Mouse Visium (n=6)
Human Visium (n=5)
Mouse scRNA (Halpern 2017)


D Pathway portal bias (raw log2 PP/PC)
Xenobiotic CYPs -0.67 -0.44 -1.21 -2.17 -2.33 -5.84
Glutamine synthesis / NH3 scavenging -1.35 -0.44 -1.37 -2.07 -2.95 -7.32
Wnt / pericentral identity -0.17 -0.03 -0.02 -1.03 -2.16 -4.04
SCFA activation (acyl-CoA synthetases) -0.14 -0.19 -0.00 -0.67 -0.30 -0.61
Bile acid synthesis -0.34 -0.16 -0.08 -0.57 -0.36 -0.35
Glycolysis -0.16 -0.01 -0.02 -0.53 0.19 0.24
Ketogenesis -0.13 0.02 0.07 -0.40 -0.33 0.05
De novo lipogenesis -0.28 -0.13 0.09 -0.37 0.57 0.26
Lactate / glycerol entry 0.25 -0.06 0.15 -0.27 -0.28 0.07
VLDL assembly / export -0.18 -0.04 0.05 -0.26 -0.12 -0.35
Mitochondrial beta-oxidation -0.02 -0.07 0.02 -0.22 -0.23 -0.12
Propionyl-CoA -> succinyl-CoA -0.00 -0.15 0.04 -0.20 -0.15 -0.08
Glucogenic amino acid catabolism 0.02 -0.13 0.10 -0.18 -0.09 -0.15
OXPHOS (complex I/III/IV/V) 0.03 -0.02 -0.06 -0.11 -0.05 0.08
Gluconeogenesis core 0.16 0.05 0.20 -0.01 0.47 0.94
Succinyl-CoA ligase / TCA entry 0.01 -0.16 -0.03 -0.00 -0.08 0.33
Urea cycle 0.20 0.12 0.21 0.09 0.46 1.03
Iron / hepcidin -0.23 0.13 0.11 0.10 0.50 0.75
Cow Stereo-seq (this study)
Cow Visium (n=2)
Pig Visium (n=5)
Human Visium (n=5)
Mouse Visium (n=6)
Mouse scRNA (Halpern 2017)
SCFA activation (acyl-CoA synthetases)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5


E Same, scaled by each dataset's |xenobiotic CYP| bias
(removes platform / resolution dynamic range)
Xenobiotic CYPs -1.00 -1.00 -1.00 -1.00 -1.00 -1.00
Glutamine synthesis / NH3 scavenging -2.02 -1.02 -1.13 -0.95 -1.27 -1.25
Wnt / pericentral identity -0.26 -0.07 -0.01 -0.47 -0.93 -0.69
SCFA activation (acyl-CoA synthetases) -0.21 -0.43 -0.00 -0.31 -0.13 -0.11
Bile acid synthesis -0.51 -0.37 -0.07 -0.26 -0.15 -0.06
Glycolysis -0.24 -0.01 -0.01 -0.25 0.08 0.04
Ketogenesis -0.20 0.04 0.05 -0.18 -0.14 0.01
De novo lipogenesis -0.41 -0.31 0.08 -0.17 0.24 0.04
Lactate / glycerol entry 0.38 -0.15 0.12 -0.12 -0.12 0.01
VLDL assembly / export -0.27 -0.09 0.04 -0.12 -0.05 -0.06
Mitochondrial beta-oxidation -0.03 -0.15 0.01 -0.10 -0.10 -0.02
Propionyl-CoA -> succinyl-CoA -0.01 -0.34 0.03 -0.09 -0.06 -0.01
Glucogenic amino acid catabolism 0.03 -0.30 0.08 -0.08 -0.04 -0.03
OXPHOS (complex I/III/IV/V) 0.04 -0.05 -0.05 -0.05 -0.02 0.01
Gluconeogenesis core 0.24 0.11 0.16 -0.01 0.20 0.16
Succinyl-CoA ligase / TCA entry 0.01 -0.36 -0.03 -0.00 -0.03 0.06
Urea cycle 0.30 0.27 0.18 0.04 0.20 0.18
Iron / hepcidin -0.34 0.31 0.09 0.05 0.22 0.13
Cow Stereo-seq (this study)
Cow Visium (n=2)
Pig Visium (n=5)
Human Visium (n=5)
Mouse Visium (n=6)
Mouse scRNA (Halpern 2017)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


F Marker and metabolic genes ('–' = below expression threshold / unmapped)
GLUL -0.8 -0.2 -1.4 -2.1 -2.8 -10.2
CYP2E1 -0.8 -0.5 -1.3 -2.1 -2.4 -6.2
CYP1A2 -0.6 -0.4 -0.2 -3.1 -2.1 -4.4
CYP7A1 -0.6 -0.4 0.4 – – -4.5
TBX3 -0.7 – – -1.6 – -4.8
RHBG -1.8 -1.5 -0.9 -2.9 – -3.4
SULT1E1 -2.2 – -0.2 – – –
HAMP 1.4 0.4 0.8 1.8 – –
GLS2 1.7 0.1 0.7 1.4 1.4 –
HAL 1.1 1.0 0.1 2.3 1.3 3.4
ALDOB 0.8 0.3 0.3 0.5 0.5 1.0
ASS1 0.4 0.2 0.2 0.6 0.4 1.3
ASL 0.2 0.1 0.0 -0.2 0.6 1.2
ARG1 0.2 -0.1 0.6 -0.0 0.5 0.9
CPS1 -0.0 0.0 0.2 -0.2 0.4 0.4
OTC 0.2 0.0 0.0 -0.6 0.2 –
PCK1 0.3 0.0 0.4 0.2 0.6 0.9
G6PC1 0.2 -0.1 -0.4 – – –
FBP1 0.2 -0.1 0.0 0.5 0.5 1.5
PC -0.2 0.3 – -0.3 -0.2 0.1
PCCA 0.1 -0.2 0.1 0.0 -0.1 0.1
PCCB -0.3 -0.2 0.2 -0.5 -0.2 -0.3
MMUT 0.0 -0.2 -0.1 -0.1 -0.3 –
HMGCS2 -0.1 0.0 – -0.2 -0.3 0.1
CPT1A -0.5 -0.2 0.1 0.0 -0.1 0.3
ACADM -0.0 -0.1 -0.1 -0.3 -0.4 -0.0
SCD -0.4 – -0.3 -0.2 – –
APOE -0.2 -0.0 0.0 -0.4 -0.3 -0.5
Cow Stereo-seq (this study)
Cow Visium (n=2)
Pig Visium (n=5)
Human Visium (n=5)
Mouse Visium (n=6)
Mouse scRNA (Halpern 2017)
portal bias (clipped ±3)
3
2
1
0
−1
−2
−3